

Chapter 12 • The Water Cycle

Water is never still. It moves endlessly between the oceans, the air, and the land in a great natural loop that scientists call the *water cycle*, or the hydrologic cycle. The same water that fills the sea today may fall on a mountain as rain next month, seep through the soil, and return to the sea through a river. Because the water only changes form and place, the Earth neither gains nor loses water. The water you drank this morning may once have floated in a cloud over a distant ocean.

Evaporation: Water Rises

The cycle is powered by the Sun. When sunlight strikes the surface of oceans, lakes, and rivers, the water warms. Some of its particles gain enough energy to escape into the air as an invisible gas called *water vapor*. This change from liquid to gas is called **evaporation**. Wet clothes drying on a line and puddles disappearing after the rain are everyday examples of evaporation at work.

Plants take part in the cycle too. Their roots draw water from the soil, and their leaves release it into the air as vapor. This process is called **transpiration**. A single large tree can release hundreds of liters of water into the air on a warm day, which is one reason forests matter so much to the climate of a region.

Condensation: Clouds Form

Warm, moist air rises. High above the ground the air is much colder, and the rising water vapor cools until it changes back into countless tiny droplets of liquid water. This change from gas to liquid is called **condensation**. The droplets gather around specks of dust and cling together, and we see them as clouds. Dew on the grass at dawn and the mist on a cold glass of water form in the same way.

Not all clouds are alike. Thin, wispy cirrus clouds float high where the air is coldest, while flat gray stratus clouds hang low and may bring drizzle. The tall, towering cumulonimbus clouds hold enormous amounts of water and bring the heavy rains and thunderstorms of the wet season. By watching the clouds, farmers and fisherfolk have long been able to guess what weather the day will bring.

DID YOU KNOW? Fog is simply a cloud that forms at ground level. When warm, moist air drifts over cool land or water during the night, the vapor condenses into tiny droplets near the surface — which is why mountain roads are often foggy at dawn.

Precipitation: Water Returns

Inside a cloud the droplets keep colliding and joining until they grow too heavy for the air to hold. Then they fall back to the Earth as **precipitation** — rain in warm regions, or snow, sleet, and hail where the air is cold. About seventy-one percent of the Earth's surface is covered by water, and most precipitation falls directly back into the sea.

The rain that falls on land takes many paths. Some of it flows downhill over the surface as **runoff**, gathering into streams and rivers that carry it back to the ocean. Some soaks into the ground — a process called **infiltration** — where it is stored between grains of soil and rock as *groundwater*. Springs and wells draw on this hidden store. And some evaporates again almost at once, beginning the journey anew.

A Cycle Without End

The water cycle has no beginning and no end. Evaporation, condensation, and precipitation follow one another continuously, driven by the energy of the Sun and the pull of gravity. The water on Earth today is the same water that existed millions of years ago, used again and again.

People depend on every stage of the cycle. Farmers wait for the rains, cities draw drinking water from rivers and wells, and forests keep the soil from washing away. Protecting watersheds and planting trees helps the cycle continue to supply the fresh water that every living thing needs.

Questions to Think About

1. What provides the energy that drives the water cycle? 2. Name the three main stages of the cycle in order. 3. What is the difference between runoff and infiltration? 4. Why is the amount of water on Earth today the same as long ago?